

Quantum Models for Corner Detection & Keypoint Extraction

项目职责与任务划分 / Task Breakdown

项目目标 / Project Goal

中文：

本项目要做一个简单可运行的量子角点/关键点检测原型。输入一张图像或局部 patch，模型判断中心位置是否是 corner / junction / keypoint。重点不是证明量子模型全面超过经典方法，而是展示低深度 QNN 能否作为一个小型局部特征分类器，并与 Harris、FAST、ORB、MLP 进行对比。

English:

This project builds a simple working prototype for quantum corner/keypoint detection. Given an image or a local patch, the model predicts whether the center location is likely to be a corner, junction, or keypoint. The goal is not to prove full quantum advantage, but to show whether a low-depth QNN can act as a compact local feature classifier compared with Harris, FAST, ORB, and MLP.

1. 项目统筹 / Project Coordination

中文：

负责确定项目范围、统一接口、检查进度和整合最终结果。需要避免项目范围过大，只做 patch-level corner/keypoint detection，不做完整 SLAM 或复杂 descriptor。

English:

Responsible for defining the project scope, unifying interfaces, checking progress, and integrating final results. The project should stay focused on patch-level corner/keypoint detection, not full SLAM or complex descriptor learning.

交付物 / Deliverables:

- 项目 README / Project README
 - 模块接口说明 / Module interface description
 - 最终 demo 逻辑 / Final demo flow
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2. 数据与标签生成 / Data and Label Generation

中文：

负责生成训练和测试数据。优先使用合成图形，例如棋盘格、矩形角点、L/T/X junction，再加入少量真实图像。标签可以来自合成真值，也可以来自 Harris / FAST / ORB pseudo-label。

English:

Responsible for generating training and testing data. Start with synthetic shapes such as checkerboards,

rectangle corners, and L/T/X junctions, then add a few real images. Labels can come from synthetic ground truth or Harris / FAST / ORB pseudo-labels.

交付物 / Deliverables:

- 合成图像数据 / Synthetic image data
 - 正负 patch 样本 / Positive and negative patches
 - 标签文件 / Label files
 - 数据可视化样例 / Data visualization examples
-

3. 经典特征提取 / Classical Feature Extraction

中文:

负责从 patch 中提取低维特征，作为 QNN 和 MLP 的共同输入。特征包括梯度、Hessian、structure tensor、Harris response 等。这样可以避免直接把整张图像输入量子线路。

English:

Responsible for extracting low-dimensional features from patches as shared inputs for QNN and MLP. Features include gradients, Hessian-like features, structure tensor, and Harris response. This avoids directly feeding the full image into a quantum circuit.

交付物 / Deliverables:

- `feature_extractor.py`
 - 固定维度特征向量 / Fixed-dimensional feature vectors
 - 特征归一化方法 / Feature normalization method
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4. 经典 Baseline / Classical Baselines

中文:

负责实现经典对照方法。至少包括 Harris、FAST、ORB 和 MLP。MLP 必须使用和 QNN 相同的输入特征，保证比较公平。

English:

Responsible for implementing classical baselines. At least Harris, FAST, ORB, and MLP should be included. The MLP must use the same input features as the QNN to ensure fair comparison.

交付物 / Deliverables:

- `baselines.py`
 - Harris / FAST / ORB 检测结果 / Detection results
 - MLP 训练结果 / MLP training results
 - baseline 指标表 / Baseline metric table
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5. QNN / VQC 模型 / QNN / VQC Model

中文：

负责实现核心量子模型。建议使用 4–8 qubit 的低深度 VQC，采用 angle encoding 或 data re-uploading。模型输出 corner probability。

English:

Responsible for implementing the core quantum model. A 4–8 qubit low-depth VQC is recommended, using angle encoding or data re-uploading. The model outputs a corner probability.

交付物 / Deliverables:

- `qnn_model.py`
 - `train_qnn.py`
 - QNN 训练曲线 / QNN training curves
 - qubit 数、depth、参数量统计 / Qubit count, circuit depth, and parameter count
-

6. 噪声鲁棒性实验 / Noise Robustness Experiments

中文：

负责测试模型在噪声、模糊、低光、低分辨率下的表现。比较 Harris、FAST、ORB、MLP 和 QNN 的稳定性。

English:

Responsible for testing model performance under noise, blur, low light, and low resolution. Compare the stability of Harris, FAST, ORB, MLP, and QNN.

交付物 / Deliverables:

- 噪声测试脚本 / Noise testing script
 - F1 / PR-AUC / repeatability 曲线
 - 噪声条件下的对比表 / Comparison table under noisy conditions
-

7. 可视化 Demo / Visualization Demo

中文：

负责把结果做成直观展示。展示原图、噪声图、不同方法检测出的 keypoints，以及 QNN 输出的 probability map。

English:

Responsible for making the results visually understandable. Show the original image, noisy image, detected keypoints from different methods, and the QNN probability map.

交付物 / Deliverables:

- `demo.ipynb` 或 `demo_app.py`
- keypoint overlay 图 / Keypoint overlay figures
- QNN probability map

- 简单资源仪表盘 / Simple resource dashboard
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8. 报告与展示材料 / Report and Presentation

中文：

负责最终报告、PPT 和讲稿。表达上要谨慎：可以说 QNN 是一个低资源局部特征分类原型，不能说已经证明量子优势。

English:

Responsible for the final report, slides, and presentation script. The wording should be careful: the QNN can be described as a compact local feature classifier prototype, but not as proven quantum advantage.

交付物 / Deliverables:

- Final report
 - Final slides
 - Demo script
 - 结果总结表 / Result summary table
 - 局限性说明 / Limitation discussion
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最小可交付版本 / Minimum Deliverable

中文：

最小版本只需要完成：

1. 合成角点数据；
2. Harris / FAST / MLP baseline；
3. 4-qubit 或 6-qubit QNN；
4. clean 和 noisy 两组实验；
5. F1、PR-AUC、repeatability；
6. keypoint overlay 可视化；
7. 一页结论说明 QNN 能支持什么、不能支持什么。

English:

The minimum version only needs:

1. Synthetic corner data;
2. Harris / FAST / MLP baselines;
3. A 4-qubit or 6-qubit QNN;
4. Clean and noisy experiments;
5. F1, PR-AUC, and repeatability;
6. Keypoint overlay visualization;
7. One conclusion page explaining what the QNN results support and what they do not support.